PROJECT-BASED LEARNING

PHASE-I EVALUATION

DNA- Destructured Neural Agent

Team ID:**DSCPP-III-2026-T051**

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T051
Semester: 3rd
Section: J,DS-2,Cyber,H
Domain: *AI / Evolutionary Robotics*

Team Members

1. *Tanishka Mahar – 2510011047*
2. *Mansha Khati – 2510380049*
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4. *Vinamra – 2510012765*

Mentor

Ms.Megha Garg

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

Robots using fixed movement strategies may fail when facing unfamiliar obstacles or changes in the environment. This creates a need for software-based simulation of adaptive movement strategies that can improve through repeated trials without manually defining every possible situation.

Motivation

- *Fixed strategies struggle with unfamiliar environments.*
- *Manual intervention limits autonomy.*
- *Rule-based systems rely on predefined conditions.*

Target Users / Stakeholders

- *AI and robotics students.*
- *Educators and researchers.*
- *Developers exploring evolutionary navigation.*

Objectives

Project Objectives

1. *Develop a software-based virtual robot simulation.*
2. *Implement a Genetic Algorithm for movement optimization.*
3. *Integrate Data Structures and OOP concepts.*
4. *Evaluate movement strategies using fitness-based test cases.*
5. *Demonstrate adaptive navigation in varied virtual environments.*

Key Objective: *Develop a graphical simulation where a virtual robot improves its navigation strategies through repeated trials and fitness evaluation.*

Proposed Solution

Proposed Approach

1. **Input:** *Define the target, robot, and virtual environment.*
2. *Generate different candidate movement strategies.*
3. *Evaluate strategies using fitness, selection, crossover, and mutation.*
4. *Store population and fitness results.*
5. *Output: Graphical robot movement and performance across generations.*

Key Features

- *Graphical Virtual Robot*
- *Virtual Obstacle Environment*
- *Genetic Algorithm-Based Adaptation*
- *Fitness-Based Evaluation*
- *Generation-wise Visualization*

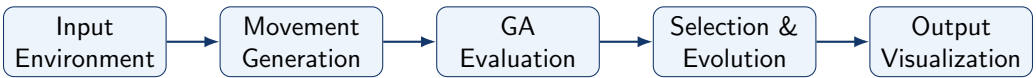
The initial GA prototype forms the core of the proposed software simulation, which will later be extended to virtual robot navigation.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, Linked Lists, Graphs and Queues*	*Environment & Navigation*
OOPs with C++	*Classes, Inheritance, Polymorphism and STL*	*Robot & GA Engine*
Operating Systems	*Not directly required in current prototype*	*Future Scope*
DBMS	*Not required; file storage can be used*	*Simulation Data*

Requirement: The project will meaningfully apply Data Structures in C and OOP concepts in C++ as part of the simulation and its core functionality.

System Architecture / Workflow



Major Components

- *Virtual Robot & Environment*
- *Genetic Algorithm Engine*
- *Visualization Module*

Data / Control Flow

- *The environment and target are provided as input.*
- *Different movement strategies are then generated and evaluated.*
- *Better-performing strategies are selected,evolved and visualized.*

Technology Stack & Methodology

Technology Stack

- Programming: *C++*
- Database: *Not Required*
- Tools / Framework: *SFML*
- IDE: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. *Requirement Analysis*
2. *System Design*
3. *GA Implementation*
4. *Graphical Simulation Development*
5. *Testing & Evaluation*

C++ supports OOP and Data Structures implementation, while SFML will provide graphical simulation and Git/GitHub will support version control.

Initial Progress & Team Contribution

Progress Achieved

- *Studied the basics of Genetic Algorithms and evolutionary robotics.*
- *Identified requirements and GA workflow.*
- *Prepared initial system architecture.*
- *Implemented the initial GA prototype.*
- *Tested fitness, selection, crossover and mutation.*

Individual Contribution

Member	Role	Contribution
Vinamra	*Lead*	*Planning & system design*
Alankrit	*Developer*	*GA prototype development*
Mansha	*Algorithm*	*Fitness, selection & evolution logic*
Tanishka	*Testing/Docs*	*Testing & documentation*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal, Design & GA Prototype
2. Phase-II: Graphical Simulation Development
3. Phase-III: Final Integration & Evaluation

Expected Outcomes

- *A working graphical virtual robot simulation.*
- *Improvement in fitness score across generations.*
- *Adaptive navigation in different virtual environments.*

References

1. *Research paper on Genetic Algorithms*
2. *SFML Official Documentation*
3. *C++ Official Documentation*

Thank You
Questions & Suggestions